

Morning Star Project

Loading Libraries

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(text2vec)  
library(Matrix)  
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
library(xgboost)
```

```
##  
## Attaching package: 'xgboost'  
  
## The following object is masked from 'package:dplyr':  
##  
##   slice
```

Loads all packages you need: data wrangling (dplyr), text features (text2vec), sparse matrices (Matrix), splitting/metrics (caret), and the model (xgboost).

Loading the data and basic cleaning

```
set.seed(42)
morningstar <- read.csv("C:/Users/YUVI/OneDrive/Desktop/MGT 599 Analysis of business strategy and capst
morningstar <- morningstar %>% filter(!is.na(ExternalName)) %>% distinct()
```

Picking the top 10 classes

```
top10 <- names(sort(table(morningstar$DetailHoldingTypeId), decreasing = TRUE)) [1:10]
top10
```

```
## [1] "E" "B" "12" "1" "BG" "BT" "ER" "BD" "CU" "BY"
```

Keep only Top 10 rows and needed columns

```
dataframe <- morningstar %>% filter(DetailHoldingTypeId %in% top10) %>%
  select(DetailHoldingTypeId, ExternalName) %>%
  na.omit()
```

Remove ultra rare BD names (<5)

```
bd_names <- dataframe %>% filter(DetailHoldingTypeId == "BD") %>% count(ExternalName, name = "n")
rare_bd <- bd_names %>% filter(n < 5) %>% pull(ExternalName)
df <- dataframe %>% filter(!(DetailHoldingTypeId == "BD" & ExternalName %in% rare_bd))
```

creating the Factor target

```
df$DetailHoldingTypeId <- factor(df$DetailHoldingTypeId)
y_fac <- df$DetailHoldingTypeId
```

Build word 1-2 gram features

```
it_word <- itoken(df$ExternalName, preprocessor = tolower, tokenizer = word_tokenizer, progressbar = TR
vocab_word <- create_vocabulary(it_word, ngram = c(1L, 2L)) %>%
  prune_vocabulary(term_count_min = 3, doc_proportion_max = 0.5, vocab_term_max = 30000)
dtm_word <- create_dtm(it_word, vocab_vectorizer(vocab_word))
```

Build Character 3 - 5 gram features

```
it_char <- itoken(df$ExternalName, progressbar = FALSE)
vocab_char <- create_vocabulary(it_char, ngram = c(3L, 5L)) %>%
  prune_vocabulary(term_count_min = 5, vocab_term_max = 15000)
dtm_char <- create_dtm(it_char, vocab_vectorizer(vocab_char))
```

Combine DTMs and apply TF-IDF

```
X_tfidf <- cbind(dtm_word, dtm_char)
tfidf <- TfIdf$new()
X <- tfidf$fit_transform(X_tfidf)
```

Split into train/valid/test

```
train_idx <- createDataPartition(y_fac, p = 0.7, list = FALSE)
X_train <- X[train_idx, ]; X_test <- X[-train_idx, ]
y_train_fac <- y_fac[train_idx]; y_test_fac <- factor(y_fac[-train_idx], levels = levels(y_train_fac))

valid_idx <- createDataPartition(y_train_fac, p = 0.2, list = FALSE)
X_valid <- X_train[valid_idx, ]; X_tr <- X_train[-valid_idx, ]
y_valid_fac <- y_train_fac[valid_idx]; y_tr_fac <- y_train_fac[-valid_idx]
```

Map factor labels to integers (0...K-1)

```
classes <- levels(y_tr_fac)
y_tr <- as.integer(factor(y_tr_fac, levels = classes)) - 1L
y_valid <- as.integer(factor(y_valid_fac, levels = classes)) - 1L
y_test <- as.integer(factor(y_test_fac, levels = classes)) - 1L
```

Build DMatrix objects and watchlist

```
dtrain <- xgb.DMatrix(data = X_tr, label = y_tr)
dvalid <- xgb.DMatrix(data = X_valid, label = y_valid)
dtest <- xgb.DMatrix(data = X_test, label = y_test)
watch <- list(train = dtrain, eval = dvalid)
```

XGboost hyperparameters

```

params <- list(objective = "multi:softprob",
  num_class = length(classes),
  eval_metric = c("mlogloss", "merror"),
  tree_method = "hist",
  max_depth = 8,
  eta = 0.2,
  subsample = 0.9,
  colsample_bytree = 0.7,
  min_child_weight = 1,
  gamma = 0)

```

```

xgb_model <- xgb.train(params = params,
  data = dtrain,
  nrounds = 1000,
  watchlist = watch,
  verbose = 1)

```

```

## [1] train-mlogloss:1.764096 eval-mlogloss:1.765408
## [2] train-mlogloss:1.492315 eval-mlogloss:1.494061
## [3] train-mlogloss:1.306473 eval-mlogloss:1.308588
## [4] train-mlogloss:1.164992 eval-mlogloss:1.168373
## [5] train-mlogloss:1.055461 eval-mlogloss:1.059296
## [6] train-mlogloss:0.970092 eval-mlogloss:0.974183
## [7] train-mlogloss:0.896386 eval-mlogloss:0.900938
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## [9] train-mlogloss:0.780746 eval-mlogloss:0.786264
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## [838] train-mlogloss:0.144370 eval-mlogloss:0.193546
## [839] train-mlogloss:0.144323 eval-mlogloss:0.193498
## [840] train-mlogloss:0.144277 eval-mlogloss:0.193468
## [841] train-mlogloss:0.144229 eval-mlogloss:0.193442
## [842] train-mlogloss:0.144182 eval-mlogloss:0.193411
## [843] train-mlogloss:0.144137 eval-mlogloss:0.193376
## [844] train-mlogloss:0.144084 eval-mlogloss:0.193319
## [845] train-mlogloss:0.144033 eval-mlogloss:0.193291
## [846] train-mlogloss:0.143978 eval-mlogloss:0.193253

```

```

## [847] train-mlogloss:0.143933 eval-mlogloss:0.193222
## [848] train-mlogloss:0.143862 eval-mlogloss:0.193190
## [849] train-mlogloss:0.143811 eval-mlogloss:0.193176
## [850] train-mlogloss:0.143763 eval-mlogloss:0.193147
## [851] train-mlogloss:0.143715 eval-mlogloss:0.193127
## [852] train-mlogloss:0.143669 eval-mlogloss:0.193095
## [853] train-mlogloss:0.143618 eval-mlogloss:0.193075
## [854] train-mlogloss:0.143571 eval-mlogloss:0.193058
## [855] train-mlogloss:0.143514 eval-mlogloss:0.193010
## [856] train-mlogloss:0.143468 eval-mlogloss:0.192977
## [857] train-mlogloss:0.143418 eval-mlogloss:0.192967
## [858] train-mlogloss:0.143366 eval-mlogloss:0.192948
## [859] train-mlogloss:0.143320 eval-mlogloss:0.192929
## [860] train-mlogloss:0.143263 eval-mlogloss:0.192911
## [861] train-mlogloss:0.143200 eval-mlogloss:0.192868
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## [864] train-mlogloss:0.143045 eval-mlogloss:0.192775
## [865] train-mlogloss:0.142990 eval-mlogloss:0.192756
## [866] train-mlogloss:0.142941 eval-mlogloss:0.192728
## [867] train-mlogloss:0.142896 eval-mlogloss:0.192698
## [868] train-mlogloss:0.142852 eval-mlogloss:0.192667
## [869] train-mlogloss:0.142808 eval-mlogloss:0.192649
## [870] train-mlogloss:0.142762 eval-mlogloss:0.192620
## [871] train-mlogloss:0.142718 eval-mlogloss:0.192592
## [872] train-mlogloss:0.142670 eval-mlogloss:0.192552
## [873] train-mlogloss:0.142623 eval-mlogloss:0.192535
## [874] train-mlogloss:0.142574 eval-mlogloss:0.192527
## [875] train-mlogloss:0.142528 eval-mlogloss:0.192510
## [876] train-mlogloss:0.142480 eval-mlogloss:0.192460
## [877] train-mlogloss:0.142431 eval-mlogloss:0.192440
## [878] train-mlogloss:0.142379 eval-mlogloss:0.192414
## [879] train-mlogloss:0.142338 eval-mlogloss:0.192374
## [880] train-mlogloss:0.142294 eval-mlogloss:0.192375
## [881] train-mlogloss:0.142247 eval-mlogloss:0.192351
## [882] train-mlogloss:0.142208 eval-mlogloss:0.192327
## [883] train-mlogloss:0.142155 eval-mlogloss:0.192289
## [884] train-mlogloss:0.142105 eval-mlogloss:0.192241
## [885] train-mlogloss:0.142047 eval-mlogloss:0.192213
## [886] train-mlogloss:0.141999 eval-mlogloss:0.192204
## [887] train-mlogloss:0.141958 eval-mlogloss:0.192174
## [888] train-mlogloss:0.141911 eval-mlogloss:0.192139
## [889] train-mlogloss:0.141861 eval-mlogloss:0.192083
## [890] train-mlogloss:0.141818 eval-mlogloss:0.192072
## [891] train-mlogloss:0.141772 eval-mlogloss:0.192023
## [892] train-mlogloss:0.141725 eval-mlogloss:0.191998
## [893] train-mlogloss:0.141673 eval-mlogloss:0.191980
## [894] train-mlogloss:0.141626 eval-mlogloss:0.191958
## [895] train-mlogloss:0.141577 eval-mlogloss:0.191951
## [896] train-mlogloss:0.141526 eval-mlogloss:0.191939
## [897] train-mlogloss:0.141471 eval-mlogloss:0.191916
## [898] train-mlogloss:0.141427 eval-mlogloss:0.191889
## [899] train-mlogloss:0.141381 eval-mlogloss:0.191846
## [900] train-mlogloss:0.141337 eval-mlogloss:0.191811

```

```

## [901]    train-mlogloss:0.141291 eval-mlogloss:0.191778
## [902]    train-mlogloss:0.141242 eval-mlogloss:0.191752
## [903]    train-mlogloss:0.141176 eval-mlogloss:0.191718
## [904]    train-mlogloss:0.141128 eval-mlogloss:0.191688
## [905]    train-mlogloss:0.141081 eval-mlogloss:0.191670
## [906]    train-mlogloss:0.141028 eval-mlogloss:0.191647
## [907]    train-mlogloss:0.140983 eval-mlogloss:0.191619
## [908]    train-mlogloss:0.140939 eval-mlogloss:0.191608
## [909]    train-mlogloss:0.140896 eval-mlogloss:0.191598
## [910]    train-mlogloss:0.140856 eval-mlogloss:0.191572
## [911]    train-mlogloss:0.140806 eval-mlogloss:0.191545
## [912]    train-mlogloss:0.140754 eval-mlogloss:0.191504
## [913]    train-mlogloss:0.140707 eval-mlogloss:0.191501
## [914]    train-mlogloss:0.140661 eval-mlogloss:0.191452
## [915]    train-mlogloss:0.140618 eval-mlogloss:0.191451
## [916]    train-mlogloss:0.140572 eval-mlogloss:0.191436
## [917]    train-mlogloss:0.140532 eval-mlogloss:0.191413
## [918]    train-mlogloss:0.140486 eval-mlogloss:0.191403
## [919]    train-mlogloss:0.140435 eval-mlogloss:0.191374
## [920]    train-mlogloss:0.140389 eval-mlogloss:0.191351
## [921]    train-mlogloss:0.140344 eval-mlogloss:0.191324
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## [923]    train-mlogloss:0.140255 eval-mlogloss:0.191284
## [924]    train-mlogloss:0.140215 eval-mlogloss:0.191258
## [925]    train-mlogloss:0.140173 eval-mlogloss:0.191228
## [926]    train-mlogloss:0.140123 eval-mlogloss:0.191202
## [927]    train-mlogloss:0.140077 eval-mlogloss:0.191195
## [928]    train-mlogloss:0.140030 eval-mlogloss:0.191165
## [929]    train-mlogloss:0.139988 eval-mlogloss:0.191155
## [930]    train-mlogloss:0.139933 eval-mlogloss:0.191119
## [931]    train-mlogloss:0.139887 eval-mlogloss:0.191082
## [932]    train-mlogloss:0.139838 eval-mlogloss:0.191075
## [933]    train-mlogloss:0.139798 eval-mlogloss:0.191078
## [934]    train-mlogloss:0.139755 eval-mlogloss:0.191039
## [935]    train-mlogloss:0.139712 eval-mlogloss:0.190988
## [936]    train-mlogloss:0.139666 eval-mlogloss:0.190974
## [937]    train-mlogloss:0.139626 eval-mlogloss:0.190965
## [938]    train-mlogloss:0.139583 eval-mlogloss:0.190953
## [939]    train-mlogloss:0.139532 eval-mlogloss:0.190912
## [940]    train-mlogloss:0.139492 eval-mlogloss:0.190894
## [941]    train-mlogloss:0.139454 eval-mlogloss:0.190888
## [942]    train-mlogloss:0.139410 eval-mlogloss:0.190853
## [943]    train-mlogloss:0.139369 eval-mlogloss:0.190830
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## [945]    train-mlogloss:0.139283 eval-mlogloss:0.190759
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## [949]    train-mlogloss:0.139100 eval-mlogloss:0.190711
## [950]    train-mlogloss:0.139059 eval-mlogloss:0.190703
## [951]    train-mlogloss:0.139019 eval-mlogloss:0.190705
## [952]    train-mlogloss:0.138974 eval-mlogloss:0.190677
## [953]    train-mlogloss:0.138932 eval-mlogloss:0.190652
## [954]    train-mlogloss:0.138892 eval-mlogloss:0.190633

```

```
## [955] train-mlogloss:0.138845 eval-mlogloss:0.190623
## [956] train-mlogloss:0.138784 eval-mlogloss:0.190580
## [957] train-mlogloss:0.138742 eval-mlogloss:0.190551
## [958] train-mlogloss:0.138705 eval-mlogloss:0.190512
## [959] train-mlogloss:0.138662 eval-mlogloss:0.190499
## [960] train-mlogloss:0.138619 eval-mlogloss:0.190466
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## [962] train-mlogloss:0.138536 eval-mlogloss:0.190397
## [963] train-mlogloss:0.138496 eval-mlogloss:0.190365
## [964] train-mlogloss:0.138446 eval-mlogloss:0.190346
## [965] train-mlogloss:0.138403 eval-mlogloss:0.190331
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## [968] train-mlogloss:0.138274 eval-mlogloss:0.190232
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## [979] train-mlogloss:0.137839 eval-mlogloss:0.189998
## [980] train-mlogloss:0.137795 eval-mlogloss:0.190003
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## [985] train-mlogloss:0.137580 eval-mlogloss:0.189941
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## [988] train-mlogloss:0.137454 eval-mlogloss:0.189868
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## [990] train-mlogloss:0.137371 eval-mlogloss:0.189816
## [991] train-mlogloss:0.137334 eval-mlogloss:0.189817
## [992] train-mlogloss:0.137298 eval-mlogloss:0.189786
## [993] train-mlogloss:0.137263 eval-mlogloss:0.189787
## [994] train-mlogloss:0.137226 eval-mlogloss:0.189752
## [995] train-mlogloss:0.137177 eval-mlogloss:0.189737
## [996] train-mlogloss:0.137130 eval-mlogloss:0.189735
## [997] train-mlogloss:0.137091 eval-mlogloss:0.189721
## [998] train-mlogloss:0.137054 eval-mlogloss:0.189695
## [999] train-mlogloss:0.137018 eval-mlogloss:0.189660
## [1000] train-mlogloss:0.136978 eval-mlogloss:0.189647
```

Predict on the test set

```
pred_prob <- predict(xgb_model, dtest)
pred_prob <- matrix(pred_prob, ncol = length(classes), byrow = TRUE)
```

```
pred_idx <- max.col(pred_prob, ties.method = "first")
pred_fac <- factor(classes[pred_idx], levels = classes)
```

Compute metrics (Accuracy, Macro F1)

```
cm <- confusionMatrix(pred_fac, y_test_fac)
acc <- as.numeric(cm$overall["Accuracy"])

prec <- cm$byClass[, "Pos Pred Value"]
rec <- cm$byClass[, "Sensitivity"]
f1 <- 2 * (prec * rec) / (prec + rec)
macro_F1 <- mean(f1, na.rm = TRUE)
```

Combine all metrics into one summary table

```
metrics_table <- data.frame(Metric = c("Accuracy", "Macro F1", "Average Precision", "Average Recall"),
  value = c(round(acc,4),
    round(macro_F1, 4),
    round(mean(prec, na.rm = TRUE), 4),
    round(mean(rec, na.rm = TRUE), 4)))
print(metrics_table)
```

```
##           Metric value
## 1      Accuracy 0.9368
## 2      Macro F1 0.9339
## 3 Average Precision 0.9502
## 4   Average Recall 0.9193
```

Per Class detailed table

```
f1_df <- data.frame(Class = sub("^Class:\\s*", "", names(f1)),
  Precision = round(prec, 4),
  Recall = round(rec, 4),
  F1_Score = round(f1, 4),
  stringsAsFactors = FALSE)
print(f1_df)
```

```
##           Class Precision Recall F1_Score
## Class: 1         1    0.9602 0.9268  0.9432
## Class: 12        12    0.9614 0.9624  0.9619
## Class: B          B    0.9318 0.8370  0.8819
## Class: BD         BD    0.9563 0.9055  0.9302
## Class: BG         BG    0.9977 0.9962  0.9970
```

## Class: BT	BT	0.9784	0.9371	0.9573
## Class: BY	BY	0.9169	0.8004	0.8547
## Class: CU	CU	0.9943	0.9702	0.9821
## Class: E	E	0.9299	0.9777	0.9532
## Class: ER	ER	0.8753	0.8797	0.8775